CFD Post-Processing

for Impeller Passages

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1. Purpose & Scope

This procedure describes a guideline for post-processing the results obtained from a Computational Fluid Dynamics (CFD) simulation activity to a single passage of impeller.

This procedure is applicable to any rotor vaned channel of radial, mixed or axial flow centrifugal pumps.

This procedure is applicable to Nova Milanese plant and La Spezia plant.

2. Abbreviations & Acronyms

Abbreviation/ Acronym	Unit of Measurement (if applicable)	Description
LE		Leading edge of the impeller's blades
TE		Trailing edge of the impeller's blades
PS		Pressure side of the impeller's blades
SS		Suction side of the impeller's blades
D2	[mm or in]	Maximum external diameter of the impeller
D1	[mm or in]	Impeller's eye diameter
Dh	[mm or in]	Diameter of the shaft's impeller
Nq or Ns		Specific speed of the pump
Eta	[%]	Efficiency
Q	[m^3/h]	Flowrate
H	[m]	Head
NPSH		Net positive suction head
CFD		Computational Fluid Dynamics
BC		Boundary Conditions
BEP		Best Efficiency Point

3. General procedure: description

The results of the analyses are obtained from the simulations prepared using the **SOP-516** *"CFD Pre-Processor for Impeller Passage" [3].*

In this simplified model, the type of analysis is steady-state, and the processed fluid is single-phase, therefore it allows to measure the performance of the vaned channel in terms of head and hydraulic efficiency.

The pros and cons related to a model of a vaned channel inserted inside a complete pump are summarized in *Table 1*:

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PROS	CONS
• The model is simple and easy to manage in the redesign phase. • Reduction of calculation time in the simulation. • It is a good model if the flow at the impeller's inlet is regular and axial, and if the interaction phenomena between the impeller and the component placed downstream of it (for example, volute or diffuser) are negligible.	• The effects of local unevenness of the incoming flow are difficult to set, unless complicated boundary conditions are imposed on it. • It does not consider the effects of fluid dynamic interaction at the exit of the rotating impeller with the stator element (for example, volute or diffuser) placed downstream of it. • Since the presence of the disks and wear rings are not simulated, it is not possible to measure the leakage flowrate and the overall impeller's efficiency. Only the internal power can be calculated.

Table 1: pros and cons of modelling

The subsequent specific discussion has the aim to provide the user with the necessary information and recommended criteria to:

- Estimate the performance of the impeller through numerical results.
- Critically evaluate post-processing results to take appropriate corrective actions.

In *Chapter 8* are present some details regarding the use of a specific purpose tool for post-processing "CFX-Solver" and "CFX-Post" inside the environment *Ansys, Engineering Simulation Software*.[4].

4. Numerical results

4.1 *Solver monitor: state of convergence*

It is possible to examine the results obtained based on the settings described in **SOP-516** *"CFD Pre-Processor for Impeller Passage"* [3].

To view the results, it is necessary to access the Solver Manager section and select the monitored variables in the results file. An example using "Ansys CFX" tool is shown in *Chapter 8.1* of this procedure.

Since all the monitored variables undergo temporal fluctuations due to numerical instability, it is always necessary to evaluate the state of convergence. For the numerical results to be considered

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reliable, the value of the monitored variable must be brought and maintained within ±1% of its "convergence band".

An example of a variable that has moved within its "convergence band" is illustrated in *Figure 1*:

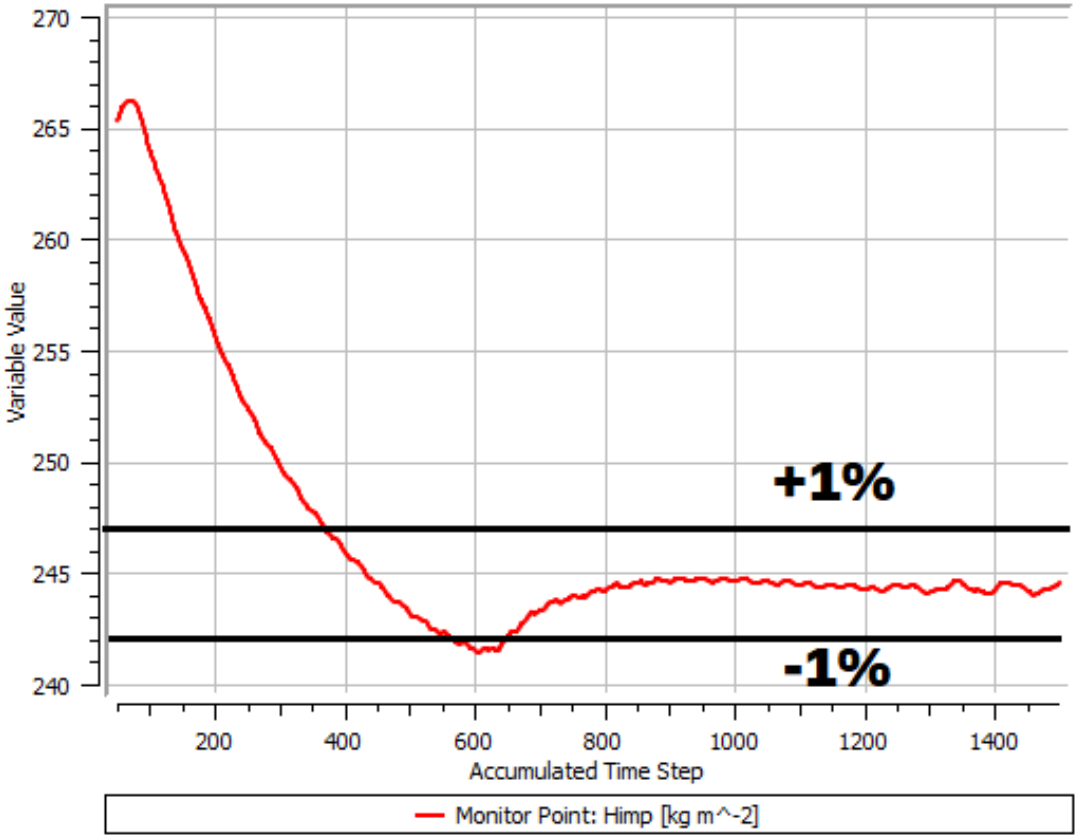

Figure 1: Monitored variable convergence

4.2 *Monitored variables*

For this type of model, the variables that have been inserted in the setting and are monitored during the solver phase are the following (refer to *Chapter 8.2* of this procedure for CFX):

$$g = 9.81 \left[\frac{m}{s^2}\right] \qquad \text{(gravitational acceleration)}$$

$$\rho = 997 \left[\frac{kg}{m^3}\right] \qquad \text{(density of water at Tamb=25°C)}$$

$$\omega = \frac{2 * \pi * n\,[rpm]}{60} \qquad \text{(angular velocity)}$$

$$n = rotational\ speed\ [rpm]$$

$$P_{min} = minimum\ pressure\ [Pa]@blade_surface$$

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$$H_{imp} = \frac{Ptot@outlet\,[Pa] - Ptot@inlet[Pa]}{g\left[\frac{m}{s^2}\right] * \rho\left[\frac{kg}{m^3}\right]}$$

$$P_{imp} = Torque@(hub, blade, shroud)[N * m] * \omega\,[s^{-1}]$$

$$ETA_{imp} = \frac{MassFlow@inlet\left[\frac{kg}{s}\right] * g\left[\frac{m}{s^2}\right] * H_{imp}[m]}{P_{imp}[W]}$$

A brief description of the used variables is presented below:

- *Pmin:* is the minimum local pressure detected at the surface of the impeller's blade.
  This variable provides qualitative and preliminary information about the entity of the cavitation phenomenon. By measuring the local pressure and considering the relative saturation pressure of the fluid in its thermodynamic conditions, it is possible to predict and deduce which impeller is best suited to satisfy the required NPSH.
  A careful evaluation and quantification of the NPSH is possible only by using cavitating models.
- *Himp*: is the head obtained by the impeller.
- *Pimp*: is the power absorbed by the single impeller vane with respect to its internal hydraulic surfaces.
- *Etaimp*: is the impeller efficiency calculated considering only the power absorbed by the impeller with respect to its internal surfaces. Since the power dissipated by ventilation is omitted, its value will be higher than expected; the redesign activity will have to take this missing contribution into account.

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An example of graphical results of the four monitored variables is showcased in *Figure 2*:

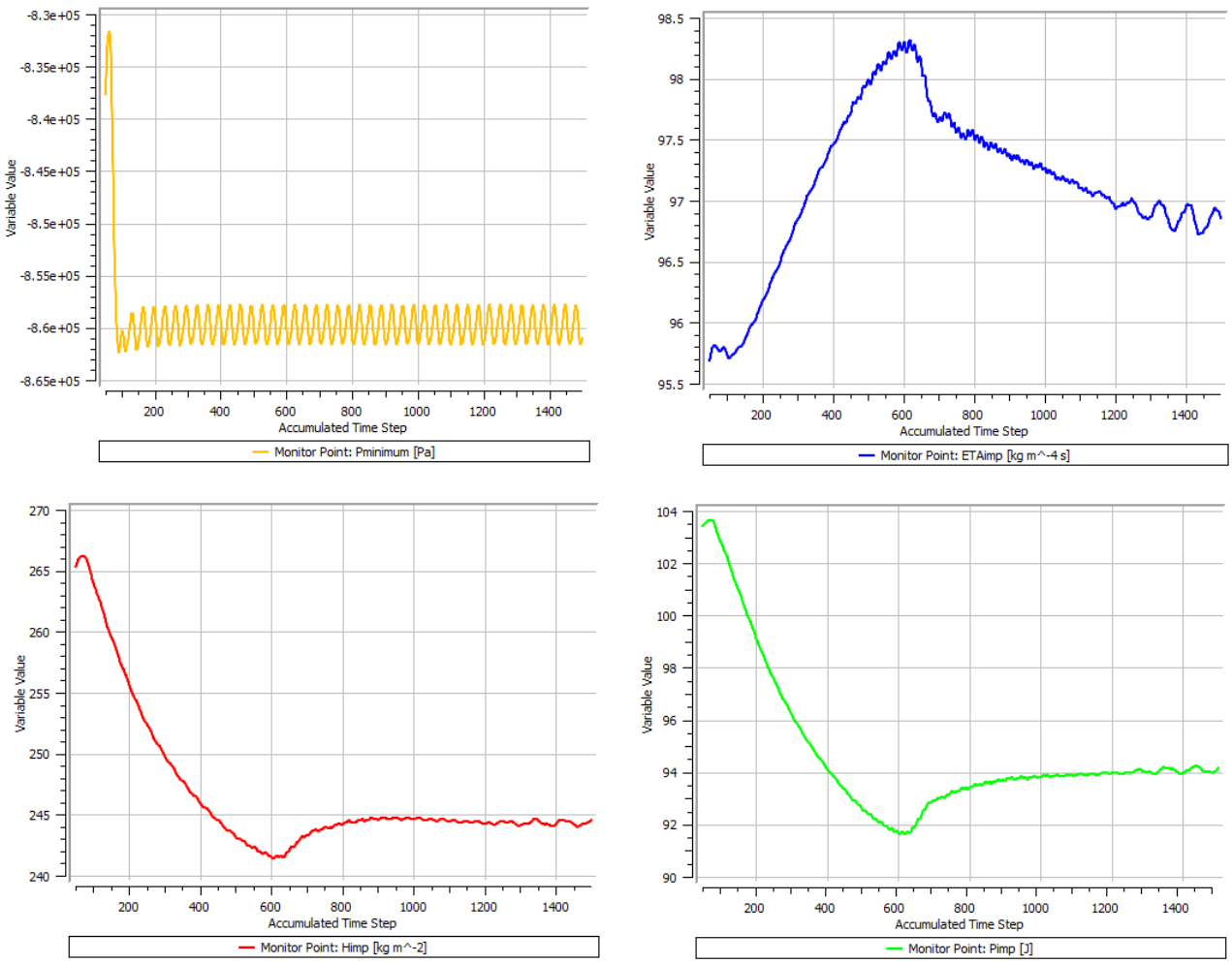

Figure 2: Graphs of monitored variables

The evaluation of these results provides immediate information in numerical terms of the impeller's performance and allows to identify the best performing ones.

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5. Graphical results

The graphical results are accessible through the Post section (refer to *Chapter 8.3* of this procedure for CFX) by selecting the results file to examine.

There are different methods for analysing the results of this model, the choice is based on the type of investigation and activity to develop; the main features are listed below:

- **Blade to Blade**
- **Meridional**
- **Blade Loading**
- **Turbo Surface at Hub Mid and Tip**
- **Streamlines on blades' surface**
- **Calculator Module**
- **Hub to Shroud Chart**
- **Report Viewer**

In addition to the expressions and variables set in **SOP-516** *"CFD Pre-Processor for Impeller Passage" [3],* it is necessary to define the following derived variables (graphical representation in *Figure 3*) to make a more detailed post and use the features previously mentioned:

- Cu: tangential velocity of the fluid in the relative reference system.
- Cm: meridian velocity of the fluid in the relative reference system.
- Cr: radial velocity of the fluid in the relative reference system.
- Alpha: angle of the fluid in the absolute reference system.

Where:

$$C_m = \sqrt{C_r^{\,2} * C_n^{\,2}}$$ (with n=axial direction of the pump)

$$Alfa = \tan^{-1}\left(\frac{C_u}{C_m}\right)$$

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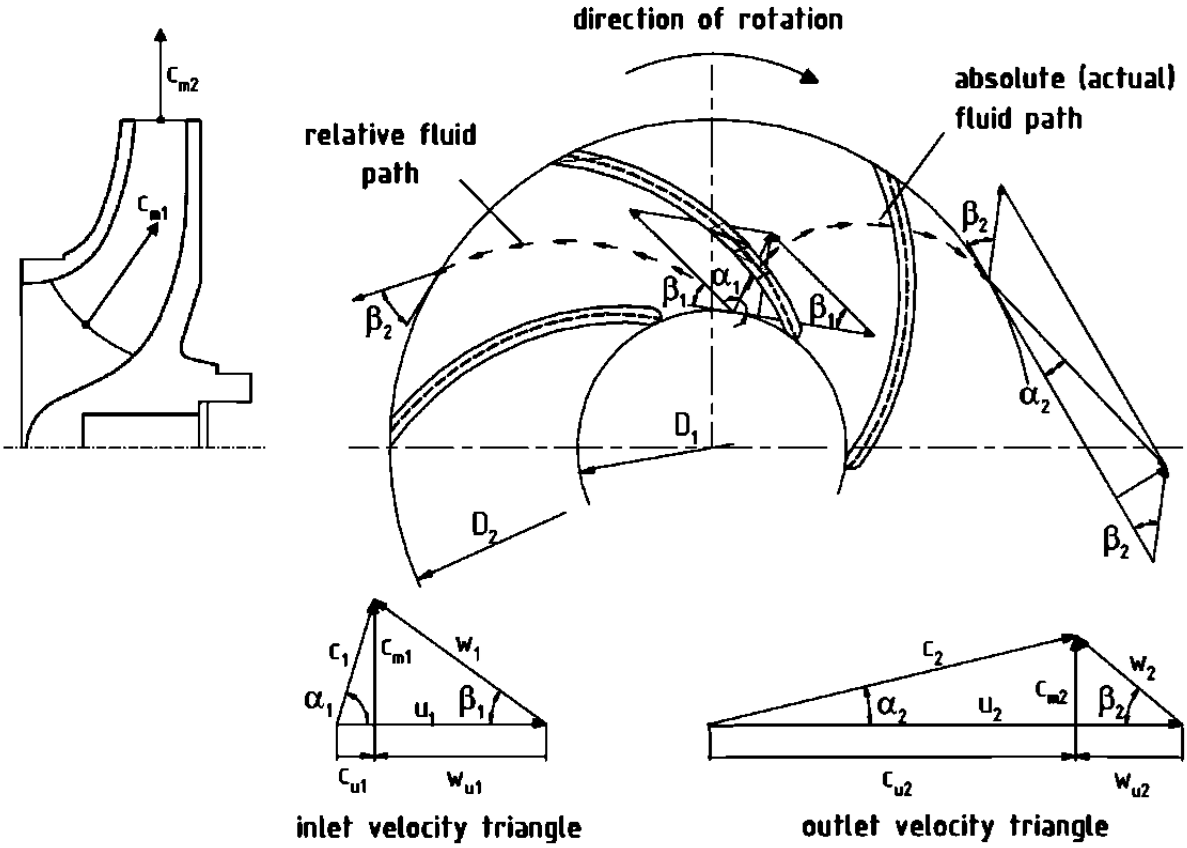

Figure 3: Characteristic velocity triangles

5.1 *Blade-to-Blade*

Activates an “unrolled annular” visualization that develops with respect to the rotation axis of the impeller. On this surface, it is possible to graph for different spans (typically values such as 0.1 for the hub, 0.5 for the mid and 0.9 for the tip) the contour of the variable of interest, the velocity vectors and the streamlines (*Figure 4*).

During the redesign activity it is important to check the pressure and velocity fields, the incidence of the flow on the LE of the blade and any possible separations and recirculation of the flow inside the blade’s vane.

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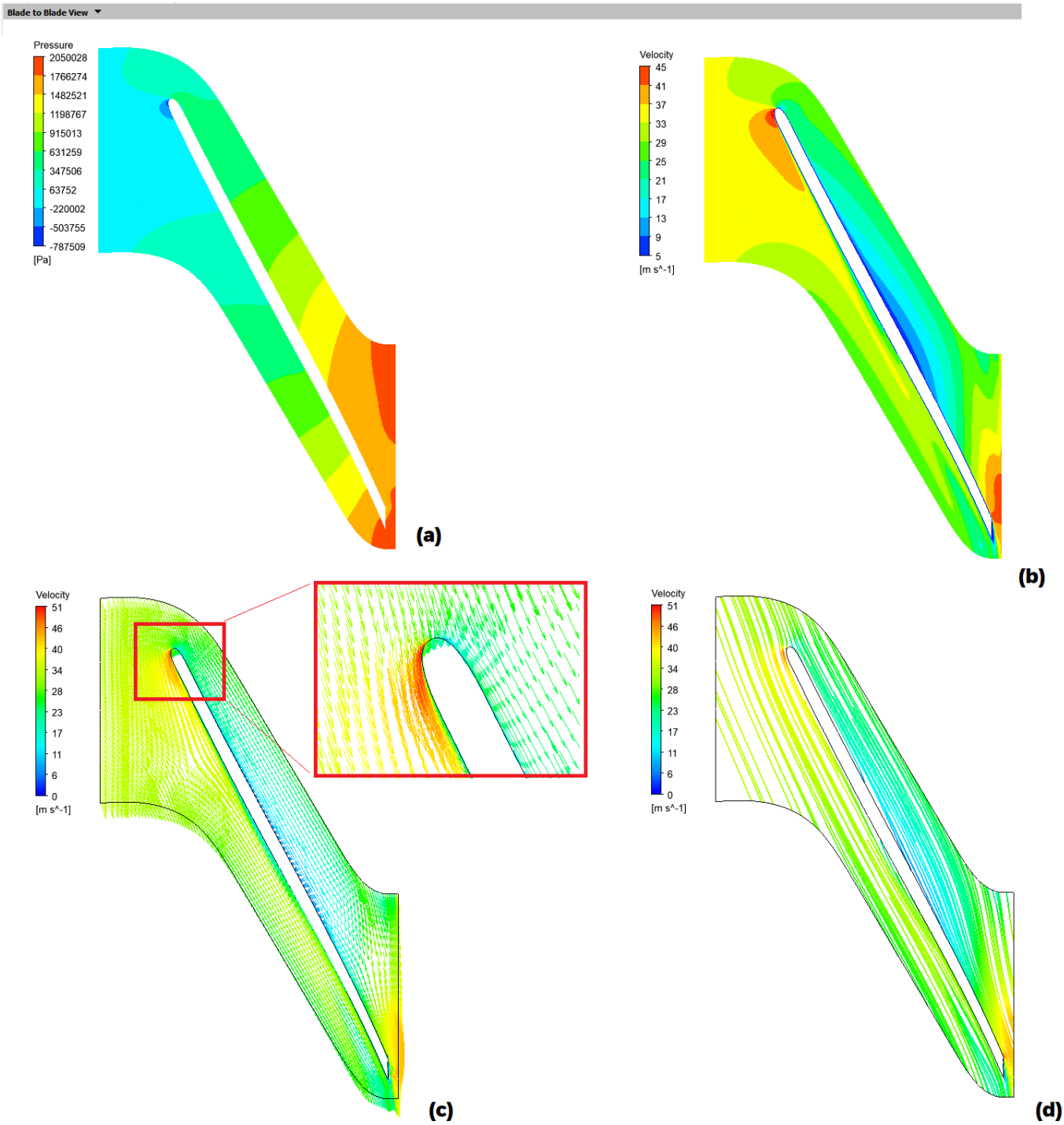

Figure 4: Plot of the Blade-to-Blade plane

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5.2 *Meridional*

Activates a meridian view of the impeller. On this surface, it is possible to plot the contour of the variable of interest and the velocity vectors. Since the impeller is axisymmetric with respect to this plane, both the variables and the velocity vectors are circumferentially averaged.

The main functions consist in verifying whether the phenomenon of “suction recirculation” is present (*Figure 5*) at the impeller inlet and whether the “velocity outward” is verified throughout the impeller outlet.

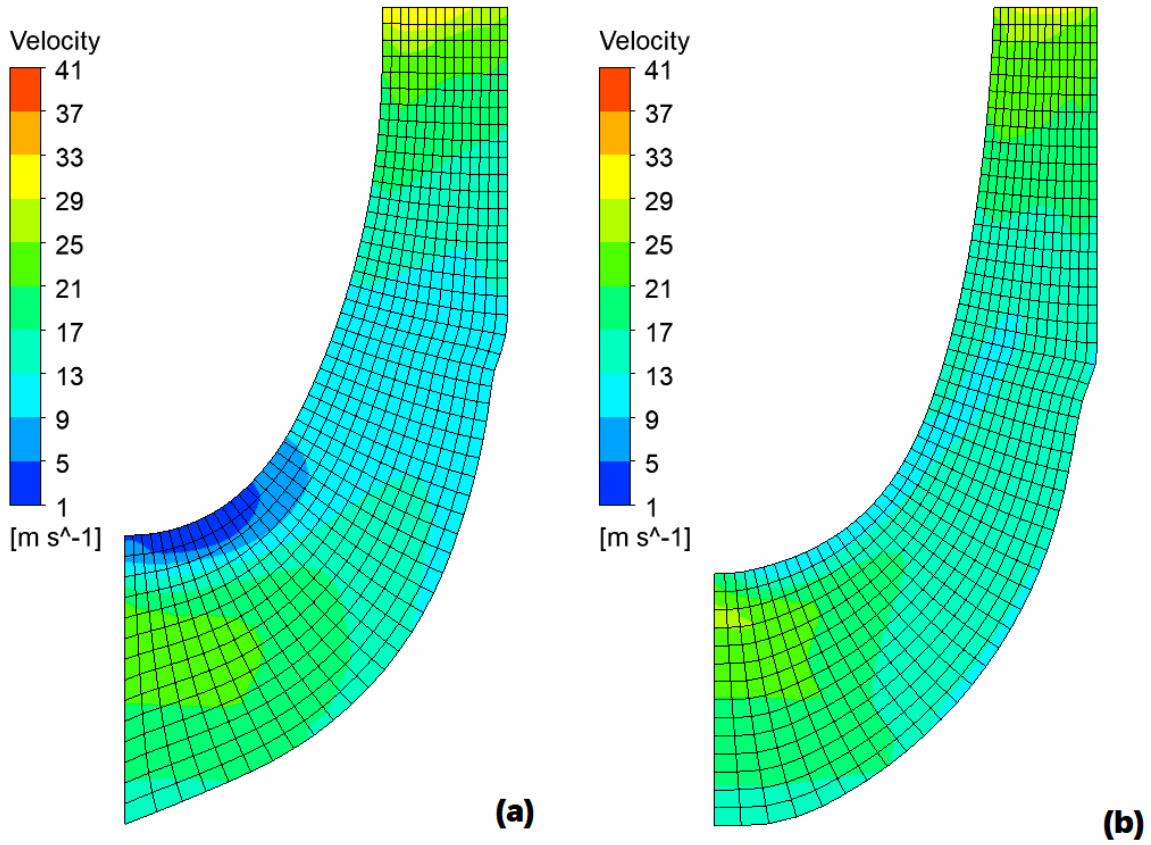

Figure 5: Plot on the meridian plane, example of “suction recirculation”

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5.3 *Blade Loading*

It shows on a graph the blade load in terms of pressure and velocity profile for different span levels and over the entire streamwise direction of the blade from LE to TE (*Figure 6*).

This feature is useful for identifying unusual velocity patterns and an irregular shape of the pressure profile.

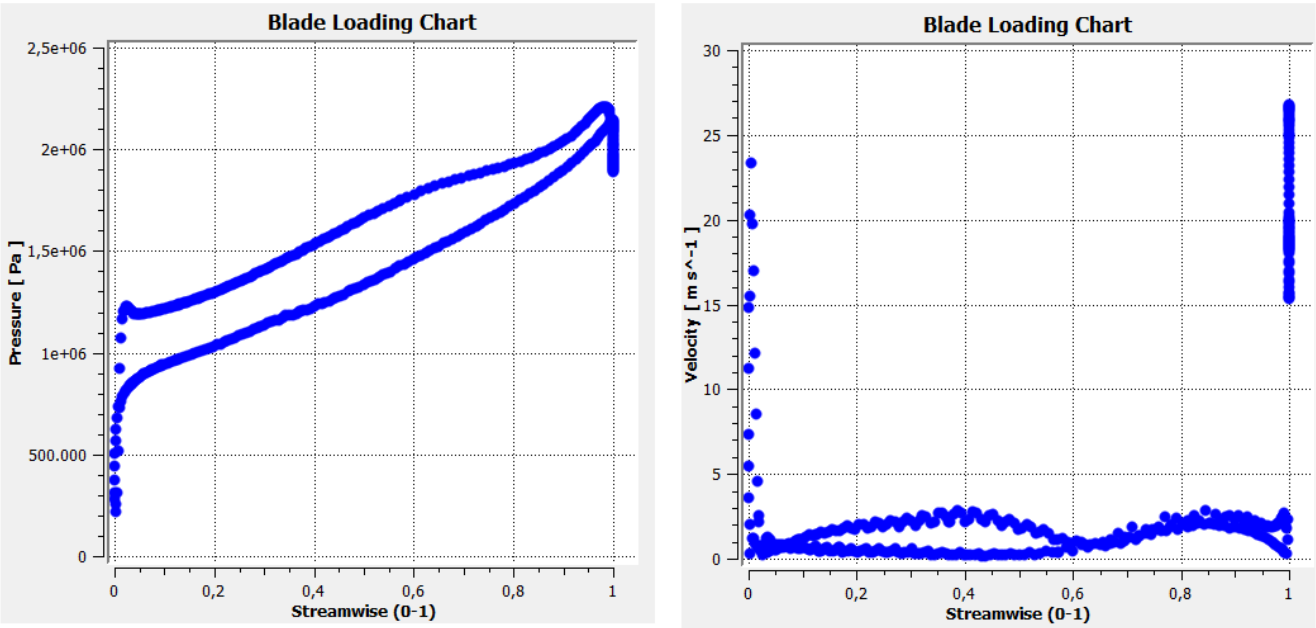

Figure 6: Plot Blade Loading

5.4 *Turbo Surface at Hub Mid and Tip*

Inside the impeller's domain, it is possible to generate surfaces called "Turbo-Surfaces".

They are "trumpet-shaped" surfaces that follow the shape of the meridian channel. These irregular surfaces can be generated for different span levels (typical values such as 0.1 for the hub, 0.5 for the mid and 0.9 for the tip are used).

It is possible to plot variables of interest like velocity vectors and streamlines on these surfaces. The streamlines are of great interest (*Figure 7*).

By enabling the multi-vane graphic visualization, it is possible to have an overall visualization of the impeller in its real 3D dimension.

This graphical option is useful for easily identifying flow separations and anomalies in flow composition. It is typically used in addition to or instead of streamlines on the Blade-to-Blade plane.

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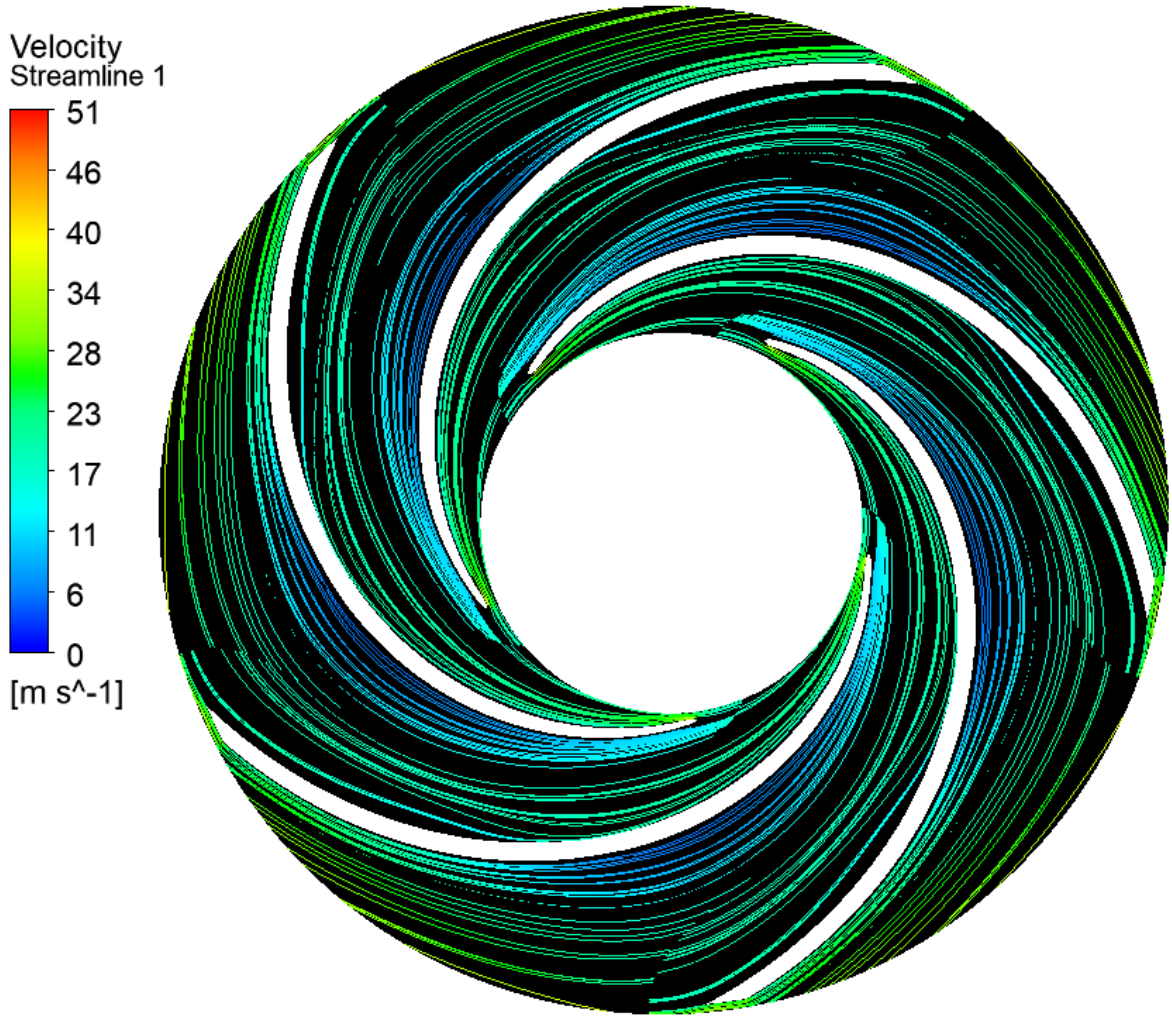

Figure 7: Plot streamlines at the Turbo-Surface

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5.5 *Streamlines on blades' surface*

This graphical feature allows, using surface streamlines, to identify and quantify the dimensions of the "secondary flow vortexes" inside the blade's compartment and of the "corner vortexes" near the LE of the blade (*Figure 8*).

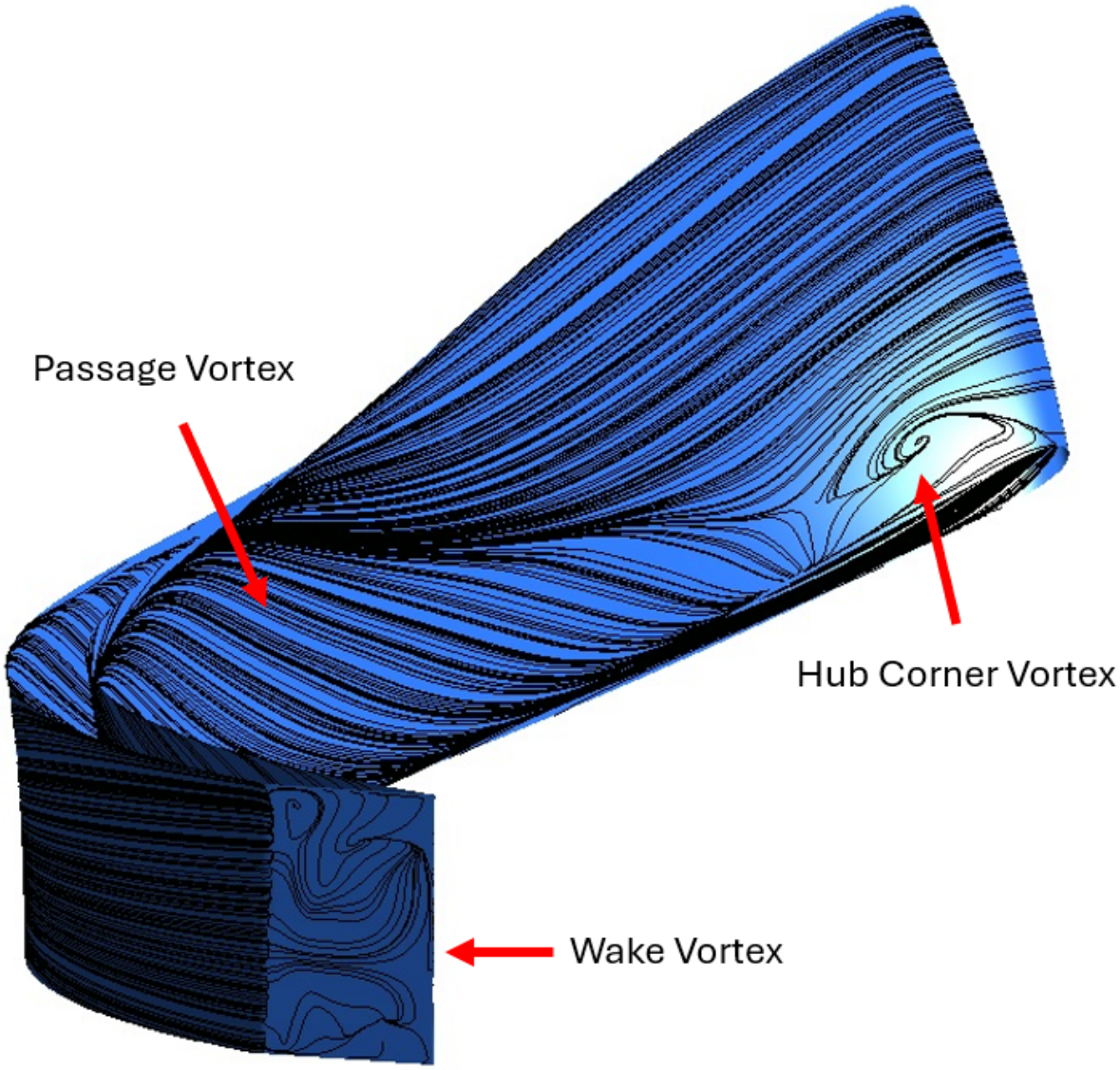

Figure 8: Plot streamlines on the Blade

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5.6 *Calculator Module*

Most post-processing tools have a specific module for calculating and quantifying variables on the different surfaces of interest.

Once the surface has been defined, whether it is created within the post or already present, it is possible to quantify the result of the variable using the appropriate functions.

Examples of functions, locations and variables useful for calculating the variable of interest are present in *Table 2* and examples of expressions with their relative results are showcased in *Table 3.*

Functions	Locations (examples)	Variables
area( )	Blade	ALFA
areaAve(<Variable|Expression>)	Hub	CM
areaInt(<Variable|Expression>)	Inlet	CTETA
areaInt_x[_<Coord Frame>]( )	Outlet	Pressure
areaInt_y[_<Coord Frame>]( )	Shroud	Radius
areaInt_z[_<Coord Frame>]( )	Plane1	Pressure
ave(<Variable|Expression>)	Plane2	Total Pressure
force( )	Isosurface1	Velocity
forceNorm [_<Axis>[_<Coord Frame>]]( )	Isosurface2	Velocity u
force_x[_<Coord Frame>]( )	…etc	Velocity v
force_y[_<Coord Frame>]( )		Velocity z
force_z[_<Coord Frame>]( )		…etc
mass()		
massAve(<Variable|Expression>)		
massFlow()		
massFlowAve(<Variable|Expression>)		
massFlowInt(<Variable|Expression>)		
massInt(<Variable|Expression>)		
maxVal(<Variable|Expression>)		
minVal(<Variable|Expression>)		
torque( )		
torque_x[_<Coord Frame>]()		
torque_y[_<Coord Frame>]()		
torque_z[_<Coord Frame>]()		
volume( )		
volumeAve(<Variable|Expression>)		
volumeInt(<Variable|Expression>)		

Table 2: Examples of functions in calculators

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Examples of expression and relative results
areaAve(ALFA)@outlet = 78.642 [degree]
areaAve(CM)@inlet = 12.6678 [m s^-1]
minVal(Pressure)@blade = -861060 [Pa]
massFlow()@inlet = 38.0204 [kg s^-1]
areaInt_z(Pressure)@blade = 990.216 [N]
…etc

Table 3: Examples of expressions and relative results

For more details about the use of the calculator module, it is preferable to consult the respective instruction manual.

For the use of the calculator module in “CFX-Post”, see the example in *Chapter 8.5* of this procedure and consult *Ansys User Manual [2].*

5.7 *Hub to Shroud Chart*

Shows plots of variables of interest against the spanwise direction (from Hub to Shroud) and against arbitrary fixed values of the streamwise (refer to *Figure 9* and *Figure 10*).

The variables of greatest interest to plot are the meridian velocity Cm, the tangential velocity Cu, the angle of the fluid in the absolute reference system Alpha and the total pressure Ptot.

This feature is useful to be able to quantify and extract the values of the variables that are plotted in the meridian channel of the Meridional feature.

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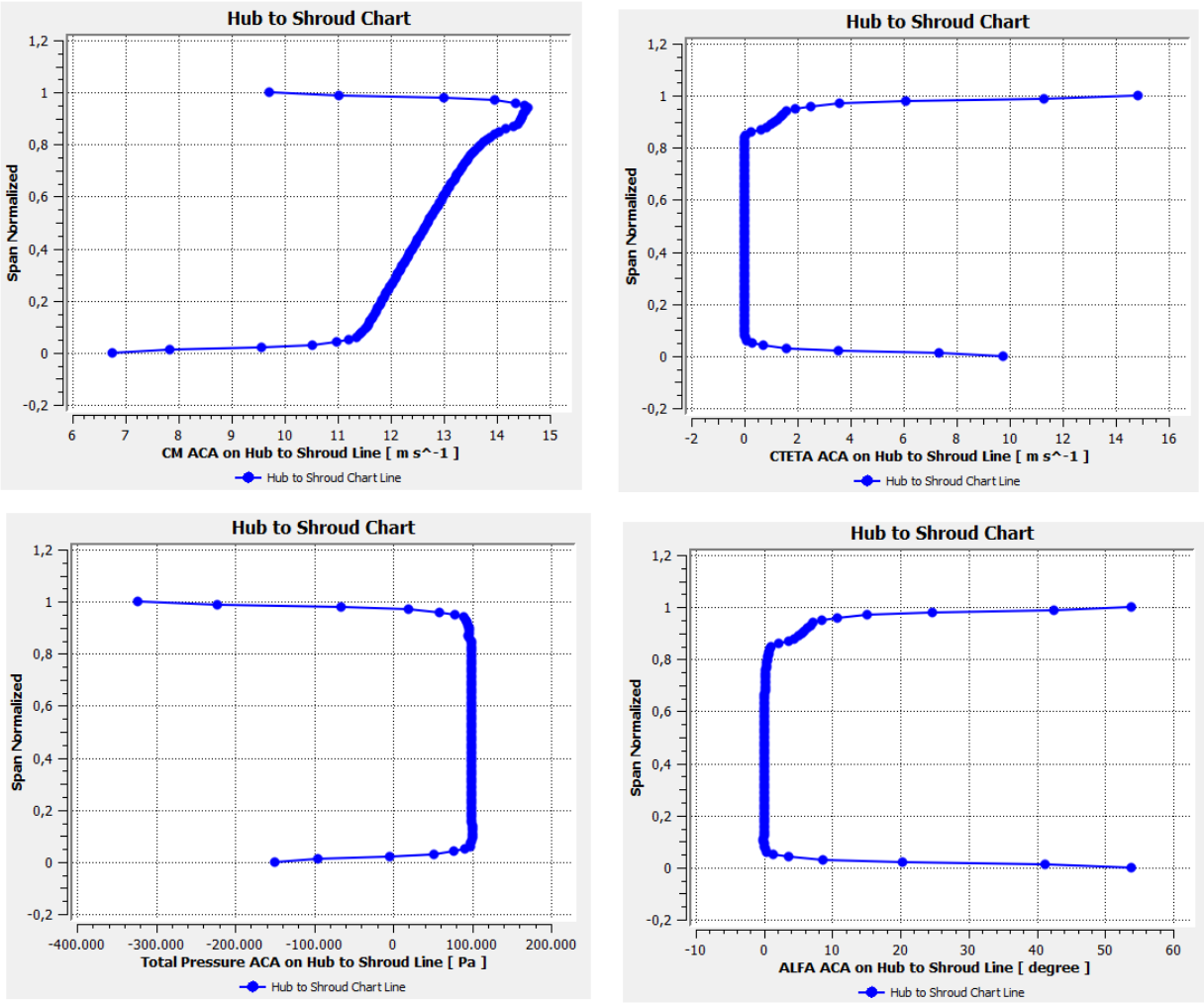

Figure 9: Hub to Shroud Chart at 0.1 streamwise for Cm (top left), Cu (top right), Ptot (bottom left) and Alfa (bottom right)

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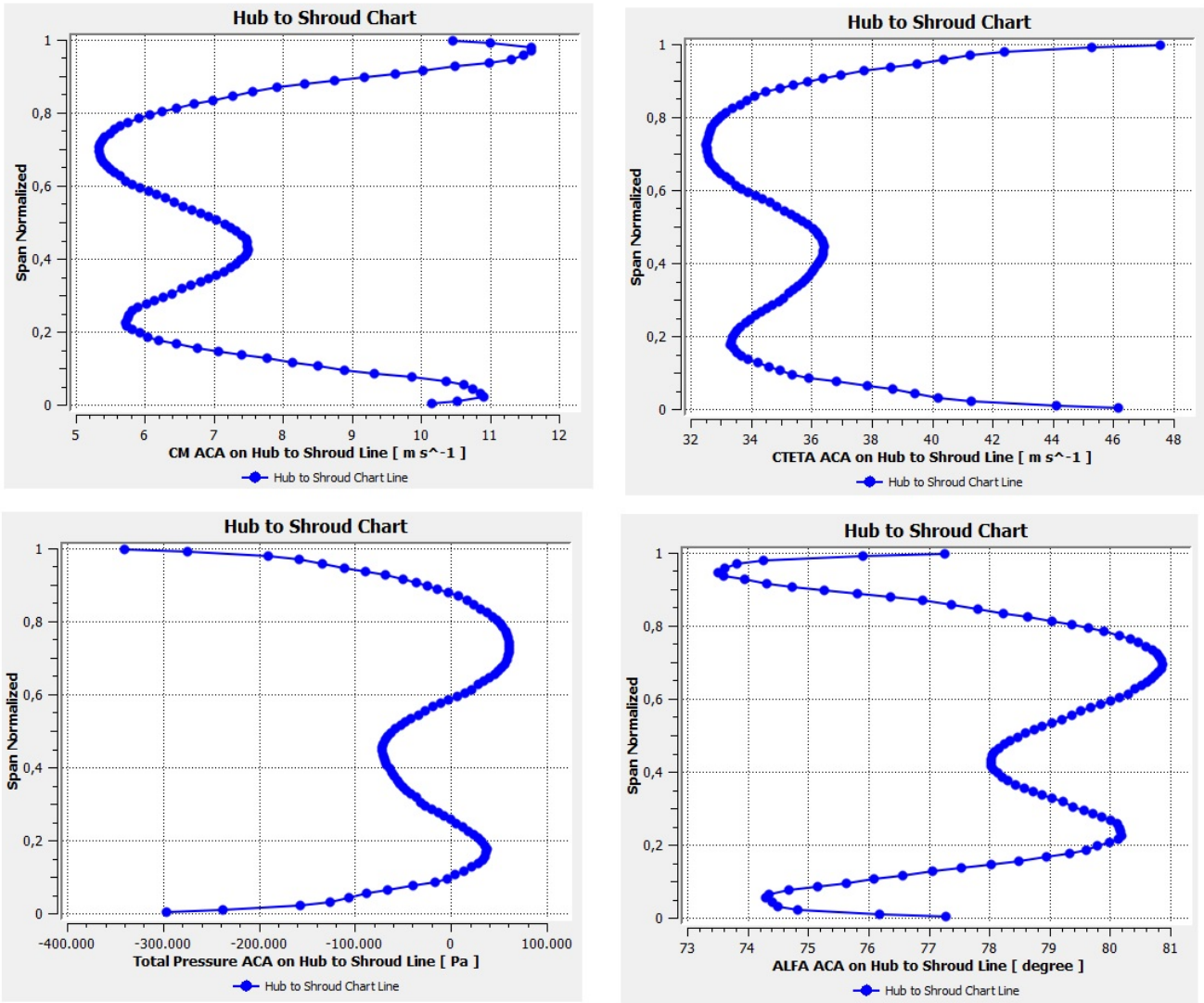

Figure 10: Hub to Shroud Chart at 0.99 streamwise Cm (top left), Cu (top right), Ptot (bottom left) and Alfa (bottom right)

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5.8 *Report Viewer*

The "Report Viewer" feature allows to save a report of the post analysis performed in the form of images and tables.

If detailed figures and tables with post-processing data have been created, it is possible to save a standardized report that collects all the post investigations processed during this activity.

In general, a standardized report contains the following paragraphs:

- Report File: the name of the Patch File, the date, the simulation time and the name and version of the solver used.
- Mesh Report: details on the mesh (number of nodes and elements and mesh quality).
- Physics Report: the model and the setting imposed as in ***SOP 516*** *"CFD Pre-Processor for Impeller Passage" [3].*
- Figures.
- Tables.

An example of a table present in the report is illustrated in *Figure 11*.

Table 5.

speed	720.0	rpm	
fluid density	997.0	kg/m3	
flow rate	1666.5	m3/h	is the flowrate of half pump. total flow rate is double
Total pressure@ in	149962.0	Pa	
Total pressure @out	438205.0	Pa	
head	29.5	m	(Ptout-Ptin)/(9.81*density)
torque	2117.7	Nm	on all rotating parts of half pump
abosrbed power	263.3	kW	torque*speed/60*2*pi.greco/1000 of half pump
pump efficiency	50.7	%	(density*9.81/3600*flow rate*head)/absorbed power/10

Figure 11: Example of a table in the report

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6. Limitations

This procedure is applicable to impellers of any geometry (radial, semi-axial and axial); furthermore, it could be used for closed, semi-open and open impellers.

This simplified model is acceptable in terms of accuracy only if it is simulated with operating conditions within the range of 0.8 to 1.1 with respect to the position of the BEP.

According to *Gulich "Centrifugal Pumps" ,4th.ed.* [1], the results of Head, Efficiency and Power in off-design operating conditions show differences between 6-8% compared to those of the test. The redesign activity where it is wanted to manage the slope of the H-Q curve is not reliable in this situation.

Gulich "Centrifugal Pumps" ,4th.ed. [1] indicates that, while using this simplified model during the redesign phase, a change greater than 2% of efficiency has a good chance of being captured by the CFD; whereas a change lower than 0.2% is insignificant since it falls within the numerical simulation error.

The setting proposed in this procedure refers to the company's background on CFD analysis using a specific software [4] as reference. These settings can be different when using different CFD software.

7. References

[1] Gulich "Centrifugal Pumps" ,4th.ed.

[2] Ansys User Manual.

[3] SOP 516 "CFD Pre-Processor for Impeller Passage".

[4] Ansys, Engineering Simulation Software.

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8. Attachments

Below are some details about the Ansys CFX solver.

8.1. Access to the control panel

To view the numerical results in Ansys CFX, you need to access the CFX-Solver Manager section (*Figure 12*) and select the monitored variables in the results file.

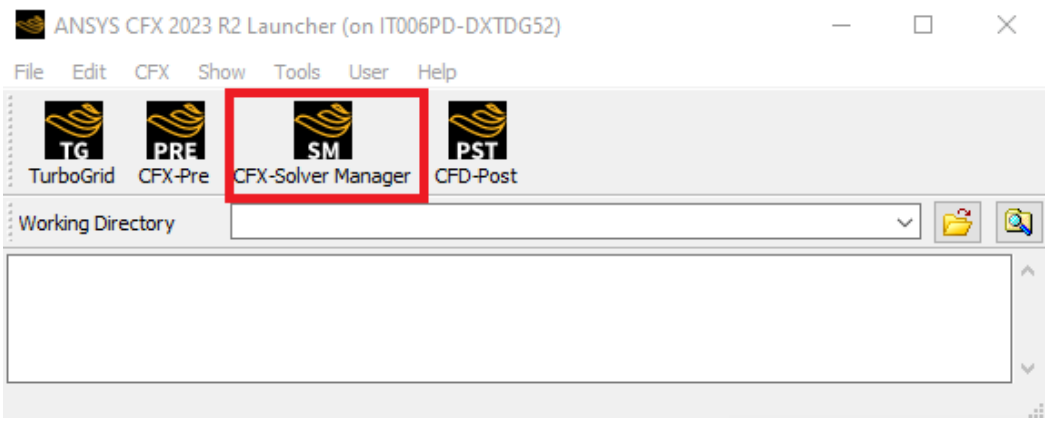

Figure 12: Control Panel, CFX-Solver Manager

8.2. Expressions of monitored variables.

Figure 13: Expressions of monitored variables

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8.3. Access to graphical results

To access the graphical results, you need to enter the “CFD-Post” section and select the results file you want to examine.

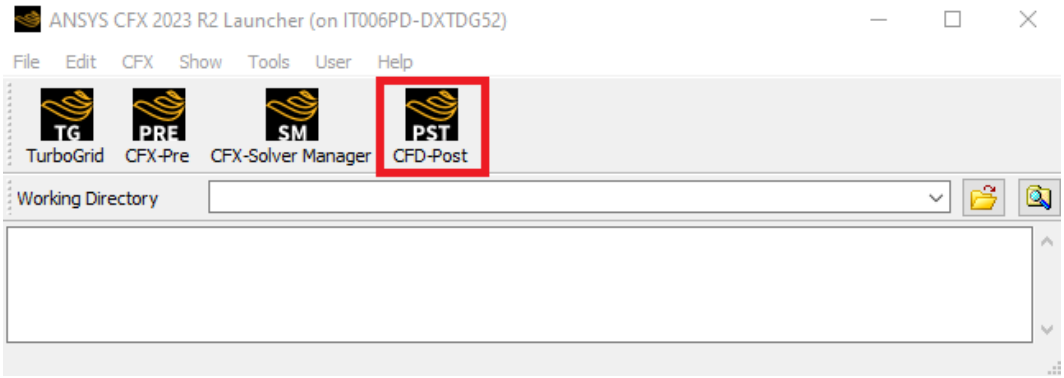

Figure 14: Control Panel, CFD-Post

8.4. Turbo module initialization

In the “CFX Turbo” module, you must first initialize the impeller domain as shown in *Figure 15*; it is possible to enable the features within the “Plots” section only once this operation has been performed.

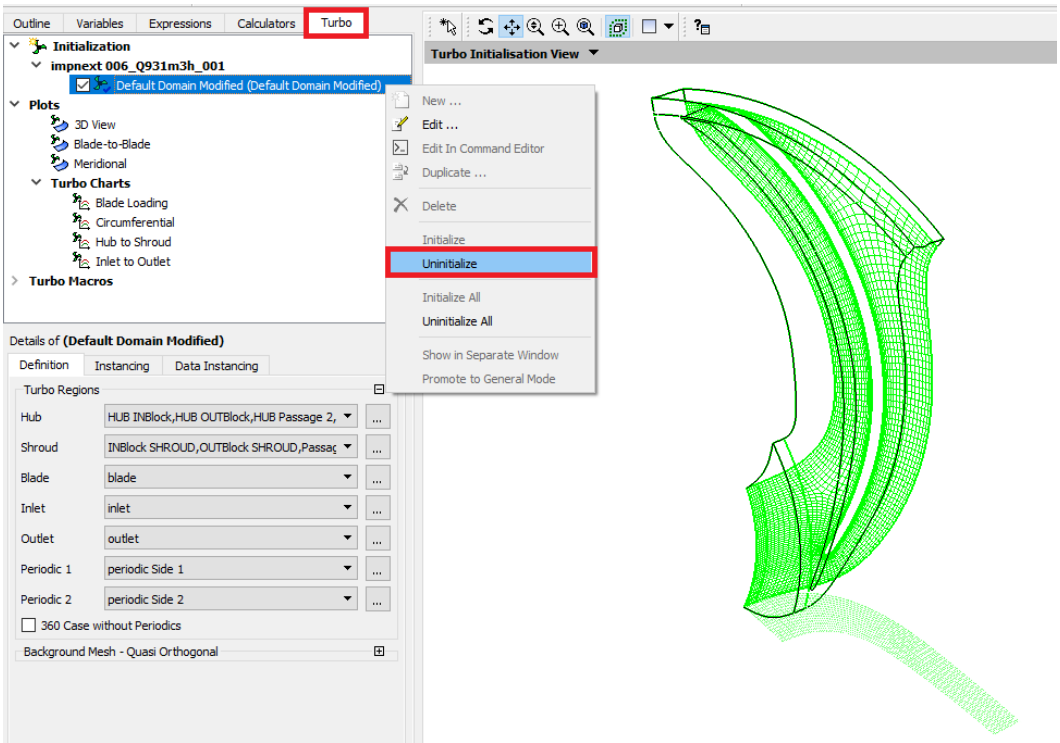

Figure 15: Turbo module and initialization

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8.5. Calculator Module

Below are shown the variables and the related expressions to be inserted inside the “Expression” and “Variables” sections in “CFX-Post” based on the type of rotation axis adopted.

```
Rotation Axis= z
Cteta=(Velocity in Stn Frame u*Y-Velocity in Stn Frame v*X)/raggio
Cr=(Velocity in Stn Frame u*X+Velocity in Stn Frame v*Y)/raggio
Cm=(Cr^2+Velocity in Stn Frame w ^2)^0.5
raggio=(X^2+Y^2)^0.5
Alfa= atan (Cteta/Cm)

Rotation Axis= x
Cteta = (Velocity in Stn Frame w*Y-Velocity in Stn Frame v*Z)/raggio
Cr = (Velocity in Stn Frame w *Z+Velocity in Stn Frame v *Y)/raggio
Cm = (Cr^2+ Velocity in Stn Frame u ^2)^0.5
raggio = (Z^2+Y^2)^0.5
alfa = atan(Cteta/Cm)

Rotation Axis= y
Cteta = (Velocity in Stn Frame u*Z-Velocity in Stn Frame w*X)/raggio
Cr = (Velocity in Stn Frame u *X+Velocity in Stn Frame w *Z)/raggio
Cm = (Cr^2+ Velocity in Stn Frame v ^2)^0.5
raggio = (X^2+Z^2)^0.5
alfa = atan(Cteta/Cm)
```

Figure 16: Expressions

By selecting the surface, the function and the variable of interest, it is possible to calculate its value. *Figure 17* shows an example of calculation of the alpha variable on the outlet surface; by selecting the specific function, it is possible to quantify its value. In this example, its value is averaged over the entire outlet area.

It is necessary to define an iso-surface with a diameter equal to D2 plus 2mm to properly quantify the Alpha2 of the impeller.

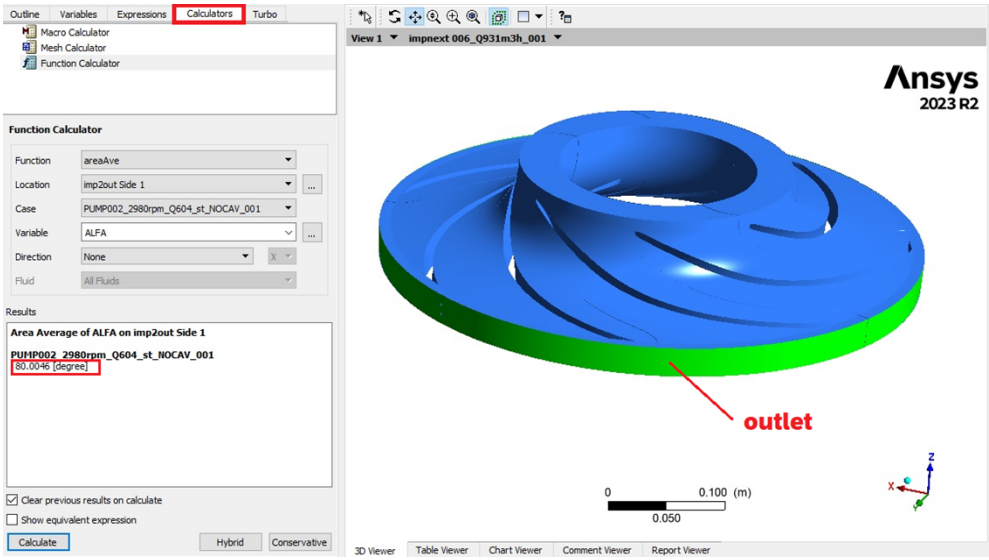

Figure 17: Calculator Module

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